



**S. B. JAIN INSTITUTE OF TECHNOLOGY, MANAGEMENT
& RESEARCH, NAGPUR.**

(An Autonomous Institute, Affiliated to RTMNU, Nagpur)



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

"To become a center for quality education in the field of computer science & engineering and to create competent professionals."

Discrete Mathematics & Graph Theory

N-PCCCS303T

TAE-II: Creative Assignment

Name of the Students	Timely Submission (1)	Originality and Content (2)	Viva (2)	Total	Signature of Course Coordinator



Family Tree as a Graph

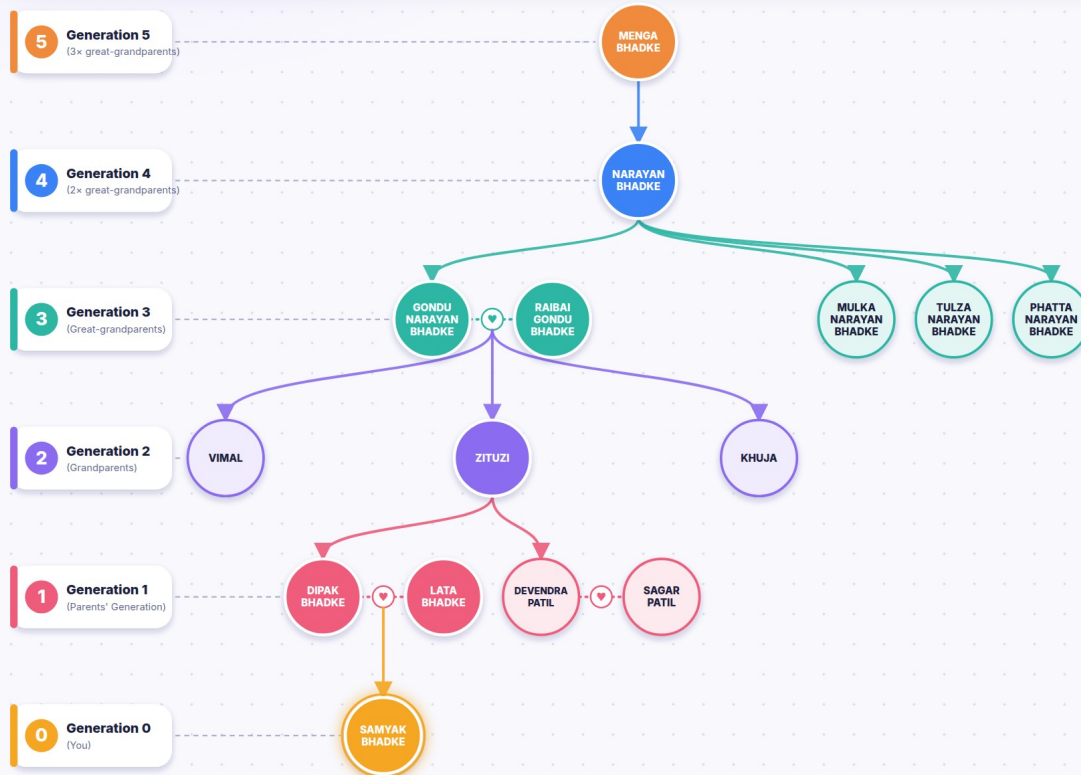
Discrete Mathematics & Graph Theory · The Bhadke Family

Submitted by: **Samyak Bhadke**

USN: **CS25041**

Submitted to: **Mama**

TAE-II DMGT



Generation Gap from You (Samyak)

PERSON	RELATION TO YOU	GEN. GAP	EDGES
Samyak Bhadke	You	0	0
Dipak Bhadke	Father	1	1
Lata Bhadke	Mother	1	2
Devendra Patil	Aunt	1	3
Sagar Patil	Uncle (by marriage)	1	4
Vimal	Grandparent's sibling	2	4
Zituzi	Grandparent	2	2
Khuja	Grandparent's sibling	2	4
Gondu Narayan Bhadke	Great-grandfather	3	3
Raibai Gondu Bhadke	Great-grandmother	3	4
Mulka Narayan Bhadke	Great-grandparent's sibling	3	5
Tulza Narayan Bhadke	Great-grandparent's sibling	3	5
Phatta Narayan Bhadke	Great-grandparent's sibling	3	5
Narayan Bhadke	2× great-grandparent	4	4
Menga Bhadke	3× great-grandparent	5	5

Shortest Path: You to Oldest Ancestor

Samyak → **Dipak** → **Zituzi** → **Gondu** → **Narayan** → **Menga**

5 edges / 5 generations

Key Observations

- Each person is a vertex (node): 15 vertices in this graph.
- Parent-child relationship is a directed edge (11); marriage is an undirected edge (3).
- Gen. gap = difference in generation level; Edges = length of the shortest path between two vertices.
- Without marriages: 12 vertices, 11 edges ($V - 1$) — connected and acyclic, so it is a tree.
- Narayan Bhadke has the most children here: 4.

Family • Relationships • Graph Theory